LAB 6

Code and Branch Coverage Analysis

Ride Share Fare application - Python test report

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SECTION	B
GITHUB LINK	github.com/Avaneesh869/SE-Labs-PES1UG24CS099/tree/main/lab6

CODE COVERAGE	BRANCH COVERAGE
100%	**100%**
19 tests passed 37 statements - 0 missing	22 tests passed 26 branches - 0 partial

Report contents

1. Code coverage Python test file
2. Code coverage terminal screenshot showing 100%
3. Branch coverage Python test file
4. Branch coverage terminal screenshot showing 100%

Prepared for Lab 6 submission.

1. Code Coverage Test File

Filename: **CodeCoverageTest_40_application.py**

```
from ride_share_fare import (
    extra_distance_fare,
    seats_and_bags_ok,
    pickup_status
)

# Tests for extra_distance_fare()
def test_unknown_cab_type():
    try:
        extra_distance_fare("bike", 10, "city", False)
        assert False
    except ValueError:
        assert True

def test_negative_distance():
    try:
        extra_distance_fare("mini", -1, "city", False)
        assert False
    except ValueError:
        assert True

def test_within_included_distance():
    assert extra_distance_fare("sedan", 5, "city", False) == 0

def test_subscriber_included_distance():
    assert extra_distance_fare("mini", 7, "city", True) == 0

def test_suv_subscriber():
    assert extra_distance_fare("suv", 13, "city", True) == 12

def test_city_fare():
    assert extra_distance_fare("mini", 10, "city", False) == 60

def test_non_city_fare():
    assert extra_distance_fare("mini", 10, "outstation", False) == 90

def test_long_distance_city():
    assert extra_distance_fare("mini", 30, "city", False) == 500

def test_long_distance_outstation():
    assert extra_distance_fare("mini", 30, "outstation", False) == 750

# Tests for seats_and_bags_ok()
```

1. Code Coverage Test File (continued)

Filename: **CodeCoverageTest_40_application.py**

```
def test_wheelchair_allowed():
    assert seats_and_bags_ok(2, 3, True) is True

def test_wheelchair_exceeds_limit():
    assert seats_and_bags_ok(3, 1, True) is False

def test_passengers_exceed_limit():
    assert seats_and_bags_ok(5, 1, False) is False

def test_bags_exceed_limit():
    assert seats_and_bags_ok(3, 3, False) is False

def test_valid_seats_and_bags():
    assert seats_and_bags_ok(4, 2, False) is True

# Tests for pickup_status()
def test_scheduled_on_time():
    assert pickup_status(10, True, False) == "on_time"

def test_unscheduled_on_time():
    assert pickup_status(5, False, False) == "on_time"

def test_airport_on_time():
    assert pickup_status(15, False, True) == "on_time"

def test_delayed_pickup():
    assert pickup_status(0, False, False) == "delayed"

def test_missed_pickup():
    assert pickup_status(-11, False, False) == "missed"
```

2. Code Coverage - Final Terminal Evidence

The final run shows 19 tests passed and 100% statement coverage (37 statements, 0 missing).

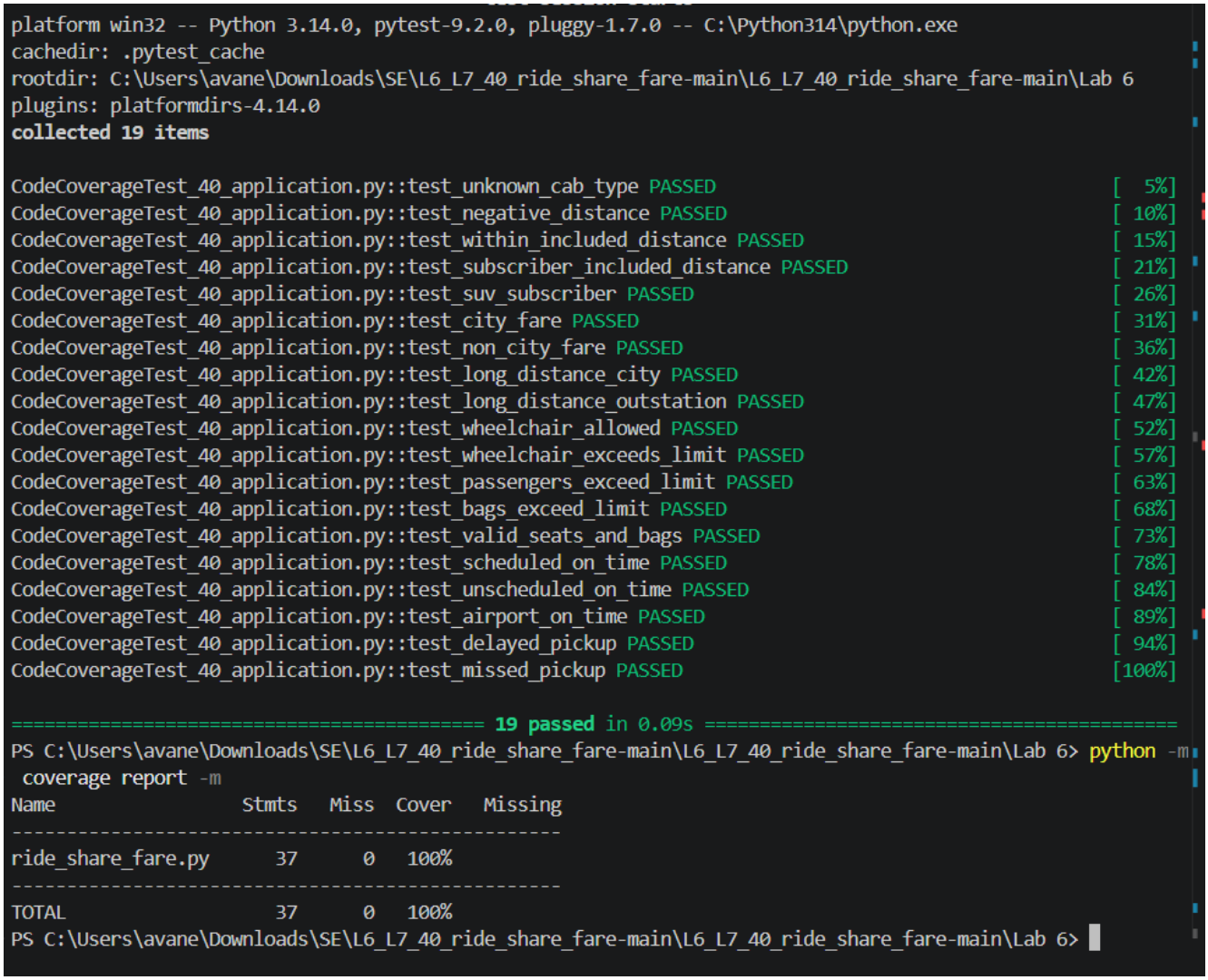


Figure 1. Terminal output for the code coverage test run.

3. Branch Coverage Test File

Filename: **BranchCoverageTest_40_application.py**

```
import pytest
from ride_share_fare import (
    extra_distance_fare,
    seats_and_bags_ok,
    pickup_status
)

# extra_distance_fare() tests
def test_unknown_cab_type():
    with pytest.raises(ValueError, match="unknown cab type"):
        extra_distance_fare("bike", 10, "city", False)

def test_negative_distance():
    with pytest.raises(ValueError, match="invalid distance"):
        extra_distance_fare("mini", -1, "city", False)

def test_subscriber_non_suv():
    assert extra_distance_fare("mini", 10, "city", True) == 24

def test_non_subscriber():
    assert extra_distance_fare("mini", 10, "city", False) == 60

def test_subscriber_suv_excluded_from_extra_km():
    assert extra_distance_fare("suv", 15, "city", True) == 36

def test_distance_at_included_limit():
    assert extra_distance_fare("mini", 5, "city", False) == 0

def test_distance_below_included_limit():
    assert extra_distance_fare("sedan", 7, "city", False) == 0

def test_city_rate():
    assert extra_distance_fare("mini", 10, "city", False) == 60

def test_non_city_rate():
    assert extra_distance_fare("mini", 10, "outstation", False) == 90

def test_extra_distance_exactly_twenty():
    assert extra_distance_fare("mini", 25, "city", False) == 240

def test_extra_distance_above_twenty_city():
    assert extra_distance_fare("mini", 26, "city", False) == 420
```

3. Branch Coverage Test File (continued)

Filename: **BranchCoverageTest_40_application.py**

```
def test_extra_distance_above_twenty_outstation():
    assert extra_distance_fare("mini", 26, "outstation", False) == 630

# seats_and_bags_ok() tests
def test_wheelchair_true_within_limit():
    assert seats_and_bags_ok(2, 5, True) is True

def test_wheelchair_true_over_limit():
    assert seats_and_bags_ok(3, 1, True) is False

def test_wheelchair_false_passenger_limit_exceeded():
    assert seats_and_bags_ok(5, 1, False) is False

def test_wheelchair_false_passenger_limit_not_exceeded():
    assert seats_and_bags_ok(4, 2, False) is True

def test_wheelchair_false_bags_exceeded():
    assert seats_and_bags_ok(3, 3, False) is False

# pickup_status() tests
def test_scheduled_on_time():
    assert pickup_status(10, True, False) == "on_time"

def test_unscheduled_on_time():
    assert pickup_status(5, False, False) == "on_time"

def test_airport_cutoff_on_time():
    assert pickup_status(15, False, True) == "on_time"

def test_delayed_pickup():
    assert pickup_status(0, False, False) == "delayed"

def test_missed_pickup():
    assert pickup_status(-11, False, False) == "missed"
```

4. Branch Coverage - Final Terminal Evidence

The final run shows 22 tests passed, 26 branches, 0 partial branches, and 100% coverage.

```
rootdir: C:\Users\avane\Downloads\SE\L6_L7_40_ride_share_fare-main\L6_L7_40_ride_share_fare-main\Lab 6
plugins: platformdirs-4.14.0
collected 22 items

BranchCoverageTest_40_application.py::test_unknown_cab_type PASSED      [  4%]
BranchCoverageTest_40_application.py::test_negative_distance PASSED     [  9%]
BranchCoverageTest_40_application.py::test_subscriber_non_suv PASSED    [ 13%]
BranchCoverageTest_40_application.py::test_non_subscriber PASSED        [ 18%]
BranchCoverageTest_40_application.py::test_subscriber_suv_excluded_from_extra_km PASSED [ 22%]
BranchCoverageTest_40_application.py::test_distance_at_included_limit PASSED [ 27%]
BranchCoverageTest_40_application.py::test_distance_below_included_limit PASSED [ 31%]
BranchCoverageTest_40_application.py::test_city_rate PASSED             [ 36%]
BranchCoverageTest_40_application.py::test_non_city_rate PASSED         [ 40%]
BranchCoverageTest_40_application.py::test_extra_distance_exactly_twenty PASSED [ 45%]
BranchCoverageTest_40_application.py::test_extra_distance_above_twenty_city PASSED [ 50%]
BranchCoverageTest_40_application.py::test_extra_distance_above_twenty_outstation PASSED [ 54%]
BranchCoverageTest_40_application.py::test_wheelchair_true_within_limit PASSED [ 59%]
BranchCoverageTest_40_application.py::test_wheelchair_true_over_limit PASSED [ 63%]
BranchCoverageTest_40_application.py::test_wheelchair_false_passenger_limit_exceeded PASSED [ 68%]
BranchCoverageTest_40_application.py::test_wheelchair_false_passenger_limit_not_exceeded PASSED [ 72%]
BranchCoverageTest_40_application.py::test_wheelchair_false_bags_exceeded PASSED [ 77%]
BranchCoverageTest_40_application.py::test_scheduled_on_time PASSED     [ 81%]
BranchCoverageTest_40_application.py::test_unscheduled_on_time PASSED   [ 86%]
BranchCoverageTest_40_application.py::test_airport_cutoff_on_time PASSED [ 90%]
BranchCoverageTest_40_application.py::test_delayed_pickup PASSED        [ 95%]
BranchCoverageTest_40_application.py::test_missed_pickup PASSED         [100%]

============================== 22 passed in 0.12s ==============================
PS C:\Users\avane\D
PS C:\Users\avane\Downloads\SE\L6_L7_40_ride_share_fare-main\L6_L7_40_ride_share_fare-main\Lab 6> python -m
 coverage report -m
Name                Stmts   Miss Branch BrPart  Cover   Missing
-----------------------------------------------------------------
ride_share_fare.py      37      0     26      0   100%
-----------------------------------------------------------------
TOTAL                   37      0     26      0   100%
PS C:\Users\avane\Downloads\SE\L6_L7_40_ride_share_fare-main\L6_L7_40_ride_share_fare-main\Lab 6>
```

Figure 2. Terminal output for the branch coverage test run.